

Contact

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Top Skills

Retrieval-Augmented Generation (RAG)

LLD

HLD

Certifications

Master Microservices with Spring Boot and Spring Cloud

Java (Basic) Certificate

Angular - The Complete Guide (2020 Edition)

The Python Bible™

Shubham Sengar

Software Development Engineer 2 @ PayPal | Java, Kubernetes, Docker, Bigtable, RAG System, MCP Server, Backend development, Distributed System, Google Cloud, SQL, NoSql, Spring Boot, Microservices, Restful Webservices

Bengaluru, Karnataka, India

Summary

At PayPal, our team developed a cutting-edge Foreign Exchange Rate Management platform, achieving a staggering efficiency gain of over 99.9%. This innovation not only streamlined onboarding for new tenants but also underpinned a robust currency trading platform that generates \$60 million in quarterly savings.

My foundation in technology stems from a BTech at the National Institute of Technology Rourkela, complemented by proficiency in Looker, MySQL, and object-oriented languages. Our work in improving authorization rates by 8% showcases my commitment to driving performance and delivering tangible business results through technical excellence and strategic problem-solving.

Key Wins:

- Implemented a Retrieval-Augmented Generation (RAG) system to estimate total trading transaction volumes with high precision.
- Implementing system optimizations that resulted in multimillion-dollar operational cost savings.
- Driving over \$1 billion in incremental revenue through advanced rate sourcing and blending strategies.

Experience

PayPal

Software Development Engineer 2

May 2022 - Present (4 years 2 months)

Bengaluru, Karnataka, India

1. Designed and implemented a Retrieval-Augmented Generation (RAG) system to estimate total trading transaction volumes by analyzing over 1,000 days of historical data, resulting in accurate volume forecasts. This RAG

system is also noted for estimating total trading transaction volumes with high precision.

2. Created a Foreign Exchange Rate Management platform to source FX rates, leveraging a blended approach with external vendors to ensure seamless execution, even in the event of scheduled process failures. This solution drastically improved the onboarding process for new tenants, reducing the timeline from three months to one hour an efficiency gain of over 99.9%.

3. Developed a multi-tenant currency trading platform enabling internal transactions between PayPal subsidiaries, including Braintree and Hyperwallet, by utilizing internal PayPal funds instead of market exchange. This solution delivers cost savings of 50 million per quarter.

4. Improved authorization rates by 8% by implementing a process to retrieve updated card details from the payment processor, reducing transaction declines.

5. Designed and implemented a gateway fee feature enabling PayPal to charge partners where contractually permitted, driving 70 million in annual revenue.

6. Enabled PayPal partners to manage the buyer fee portion of refund transactions by leading cross-functional collaboration with seven teams, including internal engineering teams, to implement API modifications. This feature has attracted 20K merchants.

Talentica Software

Software Developer

June 2021 - May 2022 (1 year)

Pune, Maharashtra, India

Designed and implemented a scalable payment platform with co-branded card integration and e-commerce capabilities, enabling merchants to seamlessly process payments to suppliers, vendors, landlords, and utility providers.

Designed and implemented high-volume integration services handling millions of daily transactions and spearheaded the end-to-end development of a rewards and cashback product from inception.

Developed a recurring batch job to track the payment status of in-flight transactions from the payment gateway and monitor recurring payment setups initiated by merchants. Implemented automated in-app and SMS notifications to keep merchants informed of status updates.

Designed and implemented the core platform for customer onboarding in the Coral Credit Card and Offers application, building the service and data processing layers from the ground up to ensure seamless functionality and scalability.

Tata Consultancy Services

Software Engineer

June 2018 - May 2021 (3 years)

Pune, Maharashtra, India

Reduced deployment times by 30% through the integration of CI/CD pipelines, enabling rapid, iterative enhancements while maintaining uninterrupted service availability.

Improved database efficiency and reliability through optimization and rigorous testing, reducing query latency by 35%.

Implemented and optimized Spring Security and RESTful APIs, significantly improving application security and data communication.

Independently analyze, troubleshoot, and rectify issues in real-time, assuming full ownership of the problem resolution process from initial diagnosis to final solution implementation.

Education

National Institute of Technology Rourkela

Bachelor of Technology - BTech · (2014 - 2018)